Project: Worked examples
Title: Conductor sizing and overload coordination
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1. Scope and method

One circuit: the design current, the smallest preferred device rating that carries it, the smallest conductor whose tabulated current-carrying capacity (your own record) carries that rating once the ambient, grouping, installation and semi-enclosed-fuse factors are applied, and the overload coordination checks I_b ≤ I_n ≤ I_z and I_2 ≤ 1.45 I_z. The context (machine wiring or installation) changes the clause citations and, for a gG fuse, MCCB or MPCB entered without I_2, whether the I_2 condition may be deemed satisfied (BS 7671 installation) or I_2 must be entered (machine wiring); every other number is the same in both contexts.

WHAT IT CALCULATES	RELATION
Design current	I_b = P/(√3·U·cosφ·η) three-phase, P/(U·cosφ·η) single-phase, P/U DC; or the linked I_b
Protective device rating	I_n = smallest preferred rating in the ratings record with I_n ≥ I_b
Fuse factor	C_c = 0.725 for a semi-enclosed fuse (1.45/2, derived), else 1
Required tabulated capacity	$I_{t,req}$ = I_n/(C_a·C_g·C_i·C_c); optional derived C_a = √[(θ_p − θ_a)/(θ_p − θ_{ref})] (derived approximation — compare with the licensed table)
Conductor cross-section	CSA = smallest cross-section in the ampacity record with I_t ≥ $I_{t,req}$; AWG gauge n: d = 0.127·92^((36 − n)/39) mm, area = π d²/4 (derived)
Carrying capacity	I_z = I_t·C_a·C_g·C_i
Device and conductor coordination	I_b ≤ I_n ≤ I_z and I_2 ≤ 1.45·I_z, I_2 = 1.45·I_n inherent for MCB types B, C, D; I_2 = 2·I_n by default for a semi-enclosed fuse; an entered I_2 (datasheet) is always checked; in the BS 7671 installation context a gG fuse to BS 88 (and an MCCB or MPCB entered without I_2, taken as I_2 ≤ 1.45·I_n) has the I_2 condition deemed satisfied when I_n ≤ I_z; in the machine-wiring context (IEC 60269-1 gG fuses ...
Device code	1 = MCB type B, 2 = type C, 3 = type D, 4 = fuse, MCCB or MPCB (carried to E5 so the fault check uses the device chosen here)

Basis: Co-ordination of conductors and overload protective devices[1](reg. 433.1.1); Current-carrying capacity of a cable[1](reg. 523.1); Current-carrying capacity of conductors and cables in normal service[2](cl. 12.4); Overcurrent protection[2](cl. 7.2); Circuit-breakers for overcurrent protection: conventional tripping current[3](clause number to be checked); Overload protection by gG fuses and semi-enclosed fuses: deemed conditions[1](reg. 433.1.201, 433.1.202).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
context	Design context (changes clause citations and the scope note only)	BS EN 60204-1 machine wiring		spec: Machine design basis MDB-2: BS EN 60204-1 wiring	any
supply	Supply	three-phase		drawing: E-SCH-021 rev A	any
P	Rated output (shaft) power; η converts it to input (DC: electrical power drawn, η not applied)	4	kW	spec: Motor schedule MS-04: conveyor drive 4 kW rated output	0.01 … 500 kW
U	Supply voltage (line voltage for three-phase)	400	V	spec: Site supply: 400 V three-phase	12 … 1000 V
$\cos_{phi}$	Power factor (not used for DC)	0.82		spec: Motor datasheet MD-04: 0.82 at full load	0.5 … 1
η	Efficiency (not used for DC)	0.85		spec: Motor datasheet MD-04: 85 % at full load	0.5 … 1
L	Circuit length (carried to the voltage-drop calculation)	35	m	drawing: E-LAY-021 rev B	0.1 … 500 m
θ_a	Ambient temperature	40	°C	spec: Machine design basis MDB-2: 40 °C inside the panel and trunking	-10 … 70 °C
insulation	Insulation	PVC 70 °C		spec: Cable schedule CS-21: PVC 70 °C single cores	any
method	Installation method key (descriptive only, e.g. B2, C, E)	B2		drawing: E-LAY-021 rev B	any
N_g	Grouped circuits	3		drawing: E-LAY-021 rev B	1 … 20
system	Conductor system (AWG: geometry only, no AWG capacities are supplied)	metric		spec: Cable schedule CS-21: metric cross-sections	any
device	Protective device	MCB type C		drawing: E-SCH-021 rev A	any
C_a	Ambient correction factor (1 does not apply the ambient: enter the factor for θ_a from your licensed table)	0.91		reference: record: owner's licensed table (not shipped); 0.91 is a synthetic value invented for this example	0.1 … 1.5

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
C_g	Grouping correction factor (at most 1)	0.77		reference: record: owner's licensed table (not shipped); 0.77 is a synthetic value invented for this example	0.1 … 1
C_i	Thermal insulation correction factor (at most 1)	1		reference: record: owner's licensed table (not shipped); 1 is a neutral value for this example	0.1 … 1
$derive_{Ca}$	Derive C_a from the temperatures instead of the entered factor	no		assumption: The entered C_a is used; the derived approximation is not applied.	any
θ_{ref}	Reference ambient temperature for the derived C_a	30	°C	assumption: Reference ambient of the licensed table; only used when C_a is derived.	0 … 60 °C
ratings	Preferred device ratings	9 row(s)		reference: record: owner's licensed table (not shipped); ratings are the typical preferred series	any
ampacity	Current-carrying capacities for the installation method	5 row(s)		reference: record: owner's licensed table (not shipped); capacities are synthetic values invented for this example	any

TABLE ROW	VALUES
ratings[1]	I_n = 6 A
ratings[2]	I_n = 10 A
ratings[3]	I_n = 16 A
ratings[4]	I_n = 20 A
ratings[5]	I_n = 25 A
ratings[6]	I_n = 32 A
ratings[7]	I_n = 40 A
ratings[8]	I_n = 50 A
ratings[9]	I_n = 63 A

TABLE ROW	VALUES
ampacity[1]	CSA = 1.5 mm²; I_t = 12.3 A
ampacity[2]	CSA = 2.5 mm²; I_t = 17.7 A
ampacity[3]	CSA = 4 mm²; I_t = 24.1 A
ampacity[4]	CSA = 6 mm²; I_t = 31.9 A
ampacity[5]	CSA = 10 mm²; I_t = 44.6 A

3. Assumptions

A1. Every tabulated value (the capacities, the device ratings, the correction factors) is entered from the user's own licensed copy or a manufacturer datasheet; none ships with the calculator. (No table from a standard or a manufacturer is shipped.)
A2. The capacities in the table are for the installation method stated; the method key is descriptive and is not interpreted. (Records carry their source and edition.)
A3. A gG fuse to BS 88 is deemed to meet $I_2 \le 1.45 \cdot I_z$ when $I_n \le I_z$ only under BS 7671 (installation context); an IEC 60269-1 gG fuse has $I_2 = 1.6 \cdot I_n$, so in the machine-wiring context I_2 is entered from the datasheet. (BS 7671 reg. 433.1.201; IEC 60269-1.)
A4. The ambient, grouping and insulation factors are applied as a product; C_g is not looked up from the number of grouped circuits. (The count is carried to the sheet and warns when $C_g = 1$ with more than one circuit.)
A5. The optional derived C_a is an approximation for the temperature difference only and is labelled as such; compare it with the licensed table. (Derived, not tabulated.)
A6. $C_c = 0.725$ applies to a semi-enclosed fuse only: 1.45 divided by an assumed fusing factor of 2. (Derived.)
A7. AWG areas follow the geometry d = 0.127·92^((36 − n)/39) mm; no AWG capacity is supplied. (Derived.)
A8. DC circuits use $I_b = P/U$; the power factor and efficiency are not applied. (Method.)
A9. derive_{Ca} (Derive C_a from the temperatures instead of the entered factor) is taken as no. (The entered C_a is used; the derived approximation is not applied.)
A10. θ_{ref} (Reference ambient temperature for the derived C_a) is taken as 30 °C. (Reference ambient of the licensed table; only used when C_a is derived.)
A11. Machine wiring context (BS EN 60204-1): the conductor is sized for the protective device rating; the capacities and factors are the user's records for the installation method stated. (The context changes citations, and whether a gG fuse, MCCB or MPCB without I_2 may be deemed to meet $I_2 \le 1.45 \cdot I_z$ (installation) or must have I_2 entered (machine wiring); no other number differs.)

4. Working

[14] $I_b = P / (\sqrt{3} * U * \cos_{phi} * \eta)$
$= 4\ \text{kW} / (\sqrt{3} \times 400\ \text{V} \times 0.82 \times 0.85)$
$= 8.2834\ \text{A}$
Design current (three-phase)

[15] I_n = smallest preferred rating with $I_n \ge I_b$ [2](cl. 7.2)
= smallest rating ≥ 8.2834 A in the ratings record
= 10 A
Rated current of the protective device (ratings record)

[16] device_{code} = 1 = MCB type B, 2 = type C, 3 = type D, 4 = fuse, MCCB or MPCB (cl. 7.2)
= MCB type C
= 2
Protective device type code (carried to E5 so the fault check uses this device)

[17] C_c = 1 (not a semi-enclosed fuse) [2](cl. 12.4)
= MCB type C
= 1
Semi-enclosed fuse factor
[21] θ_{op} = 70 degC
= 70 degC
Conductor operating temperature for the insulation (→ E5 θ, E3 T_c)
[24] $I_{t,req} = I_n / (C_a * C_g * C_i * C_c)$ [2](cl. 12.4)
= 10 A / (0.91 × 0.77 × 1 × 1)
= 14.271 A
Required tabulated capacity
[25] I_t = I_t of the selected row of the ampacity record [2](cl. 12.4)
= $I_{t,req}$ = 14.271 A
= 17.7 A
Tabulated current-carrying capacity of the selected conductor
[26] CSA = smallest cross-section in the ampacity record with $I_t \geq I_{t,req}$ [2](cl. 12.4)
= $I_{t,req}$ = 14.271 A
= 2.5 mm²
Selected conductor cross-section
[27] $I_z = I_t * C_a * C_g * C_i$ [2](cl. 12.4)
= 17.7 A × 0.91 × 0.77 × 1
= 12.402 A
Current-carrying capacity in the installed conditions
[28] $u_{Ib} = I_b / I_n$ [2](cl. 7.2; BS 7671 reg. 433.1.1)
= 8.2834 A / 10 A
= 0.82834
Design current utilisation of the device
[29] $u_{In} = I_n / I_z$ [2](cl. 7.2; BS 7671 reg. 433.1.1)
= 10 A / 12.402 A
= 0.8063
Device rating utilisation of the conductor
[30] I2 = 1.45 * I_n [2](cl. 7.2; BS 7671 reg. 433.1.1)
= 1.45 × 10 A
= 14.5 A
Conventional operating current (inherent 1.45 I_n)
[31] $u_{I2} = I2 / (1.45 * I_z)$ [2](cl. 7.2; BS 7671 reg. 433.1.1)
= 14.5 A / (1.45 × 12.402 A)
= 0.8063
I_2 utilisation of 1.45 I_z

5. Results

SYMBOL	VALUE	DESCRIPTION
I_b	8.283 A	Design current
I_n	10 A	Rated current of the device
CSA	2.500 mm²	Selected cross-section
I_z	12.40 A	Current-carrying capacity
L	35 m	Circuit length
θ_{op}	70 °C	Conductor operating temperature for the insulation
$device_{code}$	2	Protective device type code
C_c	1	Semi-enclosed fuse factor

SYMBOL	VALUE	DESCRIPTION
C_a	0.91	Ambient factor
$I_{t,req}$	14.27 A	Required tabulated capacity
I2	14.50 A	Conventional operating current
u_{Ib}	0.83	I_b / I_n (governing)
u_{In}	0.81	I_n / I_z
u_{I2}	0.81	I_2 / (1.45 I_z)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Design current within device rating ($I_b \le I_n$)	u_{Ib} = 0.83	$u_{Ib} \le 1$	0.83	²(cl. 7.2; BS 7671 reg. 433.1.1)
PASS	Device rating within conductor capacity ($I_n \le I_z$)	u_{In} = 0.81	$u_{In} \le 1$	0.81	²(cl. 7.2; BS 7671 reg. 433.1.1)
PASS	Conventional operating current within 1.45·I_z ($I_2 \le$ 1.45·I_z)	u_{I2} = 0.81	$u_{I2} \le 1$	0.81	²(cl. 7.2; BS 7671 reg. 433.1.1)

7. Applicability

WARNING records: tabulated values not taken from a reference record are ampacity (Cable current-carrying capacities); C_a, C_g, C_i (Cable correction factors); ratings (Preferred device ratings). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

Template elec.conductor-sizing@1.3.0 Content hash 549b80a7a72f70a5 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. BS EN 60204-1:2018.
3. BS EN 60898-1.